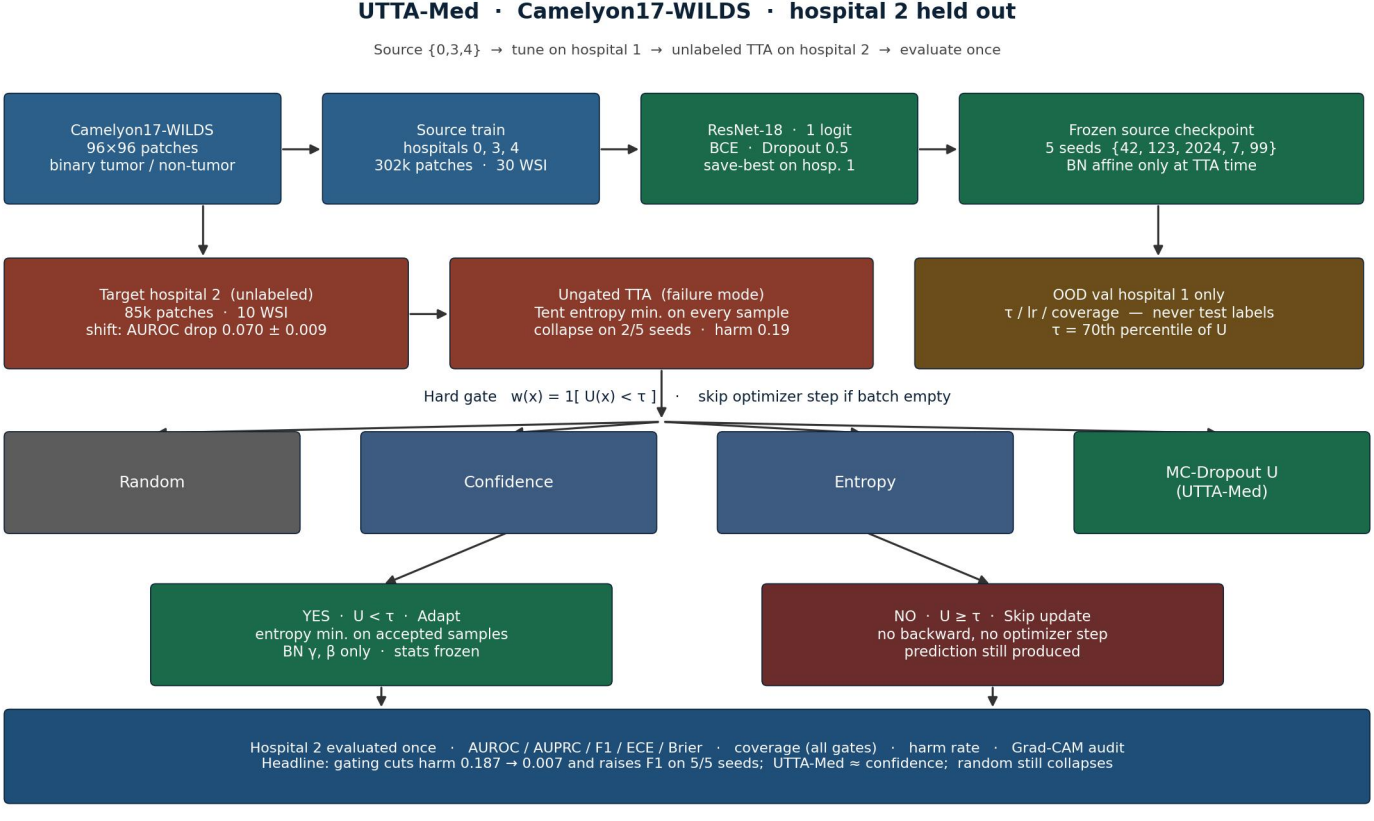

UTTA-Med · Camelyon17-WILDS · hospital 2 held out
Source {0,3,4} → tune on hospital 1 → unlabeled TTA on hospital 2 → evaluate once
Camelyon17-WILDS
96×96 patches
binary tumor / non-tumor
Source train
hospitals 0, 3, 4
302k patches · 30 WSI
ResNet-18 · 1 logit
BCE · Dropout 0.5
save-best on hosp. 1
Frozen source checkpoint
5 seeds {42, 123, 2024, 7, 99}
BN affine only at TTA time
Target hospital 2 (unlabeled)
85k patches · 10 WSI
shift: AUROC drop 0.070 ± 0.009
Ungated TTA (failure mode)
Tent entropy min. on every sample
collapse on 2/5 seeds · harm 0.19
OOD val hospital 1 only
τ / lr / coverage — never test labels
τ = 70th percentile of U
Hard gate w(x) = 1[ U(x) < τ ] · skip optimizer step if batch empty
Random
Confidence
Entropy
MC-Dropout U
(UTTA-Med)
YES · U < τ · Adapt
entropy min. on accepted samples
BN γ, β only · stats frozen
NO · U ≥ τ · Skip update
no backward, no optimizer step
prediction still produced
Hospital 2 evaluated once · AUROC / AUPRC / F1 / ECE / Brier · coverage (all gates) · harm rate · Grad-CAM audit
Headline: gating cuts harm 0.187 → 0.007 and raises F1 on 5/5 seeds; UTTA-Med ≈ confidence; random still collapses